

Case Study

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Case Study 1 — Employee Salaries

A company records the annual salaries (in ₹ lakhs) of 9 employees:

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Find the **mean, median, and mode**.
2. Find the **range**.
3. Calculate the **variance and standard deviation**.
4. Identify the outlier.
5. Which measure of central tendency best represents the typical salary? Explain why

Solution:

CASE STUDY 1: EMPLOYEE SALARIES

Given Data

A company records the annual salaries of 9 employees in ₹ lakhs:

4, 5, 5, 6, 7, 8, 8, 9, 30

The questions are:

1. Find the mean, median, and mode.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Identify the outlier.
5. Determine which measure of central tendency best represents the typical salary and explain why.

1. Calculation of Mean, Median, and Mode

A. Mean

Definition

The **mean** is the arithmetic average of all observations in a dataset. It is calculated by adding all the values and dividing the total by the number of observations.

Formula

Mean = Sum of all observations / Number of observations

The salary values are: 4, 5, 5, 6, 7, 8, 8, 9, 30

Step 1: Find the total salary

$$\text{Sum} = 4 + 5 + 5 + 6 + 7 + 8 + 8 + 9 + 30$$

$$\text{Sum} = 82$$

Step 2: Find the number of employees

Number of employees, $n = 9$

Step 3: Apply the formula

$$\text{Mean} = 82 / 9$$

$$\text{Mean} = 9.111\dots$$

Therefore,

Mean salary = ₹9.11 lakhs

Interpretation

The mean salary is approximately **₹9.11 lakhs per year**. However, this value is influenced by the salary of ₹30 lakhs. Most employees earn between ₹4 lakhs and ₹9 lakhs, but one employee earns a much higher salary. Therefore, the mean is greater than many of the individual salary values.

B. Median

Definition

The **median** is the middle value of a dataset when all observations are arranged in ascending or descending order.

The given data is already arranged in ascending order:

4, 5, 5, 6, 7, 8, 8, 9, 30

Step 1: Find the number of observations

$$n = 9$$

Since the number of observations is odd, the median is found using the following formula:

$$\text{Median position} = (n + 1) / 2$$

$$\text{Median position} = (9 + 1) / 2$$

$$\text{Median position} = 10 / 2$$

$$\text{Median position} = 5$$

Step 2: Identify the fifth value

The values can be numbered as follows:

Position	Salary in ₹ lakhs
1	4
2	5
3	5
4	6
5	7
6	8
7	8
8	9
9	30

The fifth value is 7.

Therefore,

Median salary = ₹7 lakhs

Interpretation

The median divides the data into two equal parts:

- Four employees earn less than ₹7 lakhs.
- Four employees earn more than ₹7 lakhs.

The median is not greatly affected by the extremely high salary of ₹30 lakhs. Therefore, it provides a better idea of the salary earned by a typical employee.

C. Mode

Definition

The **mode** is the value that occurs most frequently in a dataset.

The salary data is:

4, 5, 5, 6, 7, 8, 8, 9, 30

Frequency of each value:

Salary in ₹ lakhs	Frequency
4	1
5	2
6	1
7	1
8	2
9	1
30	1

The values **5** and **8** each occur two times. All other values occur only once.

Therefore,

Mode = ₹5 lakhs and ₹8 lakhs

The dataset has two modes. Hence, it is called a **bimodal dataset**.

Interpretation

The mode shows the salary values that occur most frequently. In this dataset, ₹5 lakhs and ₹8 lakhs are the most common salary levels.

Final Answer for Question 1

- **Mean = ₹9.11 lakhs**
- **Median = ₹7 lakhs**
- **Mode = ₹5 lakhs and ₹8 lakhs**
- The dataset is **bimodal** because it has two modes.

2. Calculation of Range

Definition

The **range** is the difference between the highest value and the lowest value in a dataset.

Formula

Range = Maximum value – Minimum value

From the data:

Maximum salary = ₹30 lakhs

Minimum salary = ₹4 lakhs

Calculation

Range = 30 – 4

Range = 26

Therefore,

Range = ₹26 lakhs

Interpretation

The salaries are spread over a range of ₹26 lakhs. This is a large range compared with the salaries of most employees. The high salary of ₹30 lakhs increases the range significantly. The range indicates the total spread between the lowest and highest salaries. However, it uses only two values and does not consider the distribution of all the other salaries.

3. Calculation of Variance and Standard Deviation

A. Variance

Definition

Variance measures how far the observations are spread from the mean. It is calculated by finding the squared difference between each value and the mean.

For this case study, the data is treated as the complete group of 9 employees. Therefore, the **population variance formula** is used.

Formula

Variance, $\sigma^2 = \Sigma(x - \mu)^2 / N$

Where:

- x = individual salary value
- μ = mean salary
- N = total number of employees
- Σ = sum of all values

The mean is:

$\mu = 9.11$ lakhs approximately

Calculation Table

Salary, x	$x - \text{Mean}$	$(x - \text{Mean})^2$
4	-5.11	26.12

Salary, x	x – Mean	(x – Mean) ²
5	–4.11	16.90
5	–4.11	16.90
6	–3.11	9.68
7	–2.11	4.46
8	–1.11	1.23
8	–1.11	1.23
9	–0.11	0.01
30	20.89	436.26

Step 1: Add the squared deviations

$$\Sigma(x - \mu)^2 = 512.89 \text{ approximately}$$

Step 2: Divide by the number of employees

$$\text{Variance} = 512.89 / 9$$

$$\text{Variance} = 56.99 \text{ approximately}$$

Therefore,

$$\text{Variance} = 56.99 (\text{₹ lakhs})^2$$

Interpretation

The variance is relatively high because the salary of ₹30 lakhs is much farther from the mean than the other salary values. Since variance uses squared differences, large deviations have a stronger effect on the result.

B. Standard Deviation

Definition

The **standard deviation** measures the average spread of data values around the mean. It is the square root of the variance.

Formula

$$\text{Standard deviation, } \sigma = \sqrt{\text{Variance}}$$

Calculation

Standard deviation = $\sqrt{56.99}$

Standard deviation = 7.55 approximately

Therefore,

Standard deviation = ₹7.55 lakhs

Interpretation: The standard deviation shows that the salaries have a considerable amount of variation around the mean salary of ₹9.11 lakhs.

The value of ₹30 lakhs is far from the mean and causes the standard deviation to become larger. Most of the salaries are grouped between ₹4 lakhs and ₹9 lakhs, but the high salary creates a large spread.

Final Answer for Question 3

- **Variance = 56.99 (₹ lakhs)²**
- **Standard deviation = ₹7.55 lakhs**

4. Identification of the Outlier

Definition

An **outlier** is a value that is unusually high or unusually low compared with the other observations in a dataset.

The salary data is:

4, 5, 5, 6, 7, 8, 8, 9, 30

Most salaries are between ₹4 lakhs and ₹9 lakhs. The value ₹30 lakhs is much larger than all the other values.

Therefore,

₹30 lakhs is the outlier.

Explanation

The difference between ₹30 lakhs and the next highest salary of ₹9 lakhs is:

$$30 - 9 = 21 \text{ lakhs}$$

This is a very large difference. Therefore, ₹30 lakhs is clearly separated from the main group of salary values.

The outlier affects several statistical measures:

1. It increases the mean.
2. It increases the range.
3. It increases the variance.

4. It increases the standard deviation.
5. It causes the distribution to become positively or right-skewed.

5. Best Measure of Central Tendency

The three measures of central tendency are:

1. Mean
2. Median
3. Mode

For this dataset, the **median** is the best measure for representing the typical employee salary.

Therefore,

The median salary of ₹7 lakhs best represents the typical salary.

Reason 1: The dataset contains an outlier

The salary of ₹30 lakhs is much higher than the other salaries. This value pulls the mean upward.

The mean is: ₹9.11 lakhs

However, most employees earn between ₹4 lakhs and ₹9 lakhs. Therefore, the mean gives a higher estimate of the typical salary.

Reason 2: The median is resistant to extreme values

The median is the middle value. It depends on the position of the observations rather than the size of every observation.

The median is: ₹7 lakhs

Even though the highest salary is ₹30 lakhs, the middle position remains ₹7 lakhs.

Therefore, the median is not strongly affected by the outlier.

Reason 3: The median represents the central salary

The median divides the employees into two groups:

- Half of the employees earn ₹7 lakhs or less.
- Half of the employees earn ₹7 lakhs or more.

Thus, ₹7 lakhs gives a realistic description of the central salary in the company.

Reason 4: The mode is less suitable

The modes are ₹5 lakhs and ₹8 lakhs. These values show the most frequently occurring salaries, but there are two modes. Therefore, the mode does not provide one single central value.

The mode is useful for identifying common salary levels, but it is less useful than the median for describing the overall typical salary.

Overall Conclusion

The annual salary data of the 9 employees is:

4, 5, 5, 6, 7, 8, 8, 9, 30

The calculated results are:

Statistical Measure	Result
Mean	₹9.11 lakhs
Median	₹7 lakhs
Mode	₹5 lakhs and ₹8 lakhs
Range	₹26 lakhs
Variance	56.99 (₹ lakhs) ²
Standard Deviation	₹7.55 lakhs
Outlier	₹30 lakhs
Best measure of typical salary	Median

The data contains one unusually high salary of ₹30 lakhs. This outlier increases the mean, range, variance, and standard deviation. As a result, the mean of ₹9.11 lakhs does not accurately represent the salary earned by most employees.

The median of ₹7 lakhs is the most appropriate measure because it is not significantly affected by the extreme salary. Therefore, the median provides the most reliable description of the typical employee salary.

Final conclusion: The typical annual salary is best represented by the median, which is ₹7 lakhs.

Case Study 2 — Website Response Time

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the mean and median.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Determine whether the data is likely to be **right-skewed or left-skewed**.
5. Explain why the median may be more useful than the mean.

Solution:

CASE STUDY 2: WEBSITE RESPONSE TIME ANALYSIS

Given Data

A data scientist records the response time of a website for 10 requests. The response times are measured in milliseconds (ms).

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

The questions are:

1. Calculate the mean and median.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Determine whether the data is right-skewed or left-skewed.
5. Explain why the median may be more useful than the mean.

1. Calculation of Mean and Median

A. Mean

Definition

The **mean** is the arithmetic average of all observations in a dataset. It is calculated by adding all the values and dividing the total by the number of observations.

Formula: Mean = Sum of all observations / Number of observations

The response-time data is:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

Step 1: Find the total of all response times

Sum = 120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300

Sum = 1505 milliseconds

Step 2: Find the number of observations

Number of requests, $n = 10$

Step 3: Apply the mean formula

Mean = 1505 / 10

Mean = 150.5 milliseconds

Therefore,

Mean response time = 150.5 ms

Interpretation

The mean indicates that the average response time is **150.5 milliseconds**. However, most response times are between 120 ms and 150 ms. The value of 300 ms is much larger than the other values and increases the mean.

Therefore, the mean is influenced by the unusually high response time.

B. Median

Definition

The **median** is the middle value of a dataset after arranging the observations in ascending or descending order.

The data is already arranged in ascending order:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

Step 1: Determine whether the number of observations is odd or even

The total number of observations is: $n = 10$

Since 10 is an even number, there are two middle values.

The positions of the two middle values are:

$n / 2 = 10 / 2 = 5$ and $(n / 2) + 1 = 5 + 1 = 6$

Therefore, the fifth and sixth values must be used.

Step 2: Identify the fifth and sixth values

Position	Response time in ms
1	120
2	125
3	128
4	130
5	132
6	135
7	140
8	145
9	150
10	300

The fifth value is: 132 ms

The sixth value is: 135 ms

Step 3: Calculate the median

Median = (Fifth value + Sixth value) / 2

Median = (132 + 135) / 2

Median = 267 / 2

Median = 133.5 ms

Therefore,

Median response time = 133.5 ms

Interpretation

The median indicates that the central response time is **133.5 milliseconds**.

The median divides the observations into two groups:

- Five response times are below the central position.
- Five response times are above the central position.

The value of 300 ms does not significantly affect the median because the median depends mainly on the position of the values.

Comparison of Mean and Median

Measure	Value
Mean	150.5 ms
Median	133.5 ms

The mean is greater than the median:

$$150.5 - 133.5 = 17 \text{ ms}$$

The mean is 17 ms higher than the median. This difference occurs because the response time of 300 ms pulls the mean upward.

Final Answer for Question 1

Mean response time = 150.5 ms

Median response time = 133.5 ms

2. Calculation of Range

Definition

The **range** measures the difference between the highest and lowest values in a dataset.

Formula

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

From the given data:

$$\text{Maximum response time} = 300 \text{ ms}$$

$$\text{Minimum response time} = 120 \text{ ms}$$

Calculation

$$\text{Range} = 300 - 120$$

$$\text{Range} = 180 \text{ ms}$$

Therefore,

$$\text{Range} = 180 \text{ ms}$$

Interpretation

The response times are spread over 180 milliseconds from the minimum value to the maximum value.

The large range is mainly caused by the response time of 300 ms. Without this value, the response times would be much closer together.

The range provides a simple measure of the total spread of the data. However, it considers only the smallest and largest values and does not use all the observations.

3. Calculation of Variance and Standard Deviation

A. Variance

Definition

Variance measures how much the data values are spread around the mean. It is calculated by finding the difference between each value and the mean, squaring each difference, adding the squared differences, and dividing by the total number of observations.

For this case study, the 10 recorded response times are treated as the complete dataset.

Therefore, the **population variance formula** is used.

Formula: Variance, $\sigma^2 = \Sigma(x - \mu)^2 / N$

Where:

- x = individual response time
- μ = mean response time
- N = total number of observations
- Σ = sum of the values

The mean is:

$\mu = 150.5$ ms

Step-by-Step Calculation Table

Response time, x	$x - 150.5$	$(x - 150.5)^2$
120	-30.5	930.25
125	-25.5	650.25
128	-22.5	506.25
130	-20.5	420.25
132	-18.5	342.25
135	-15.5	240.25
140	-10.5	110.25
145	-5.5	30.25
150	-0.5	0.25
300	149.5	22,350.25

Step 1: Find the sum of squared deviations

$$\Sigma(x - \mu)^2 = 25,581.50$$

Step 2: Divide by the number of observations

$$\text{Variance} = 25,581.50 / 10$$

$$\text{Variance} = 2,558.15$$

Therefore,

$$\text{Variance} = 2,558.15 \text{ ms}^2$$

Interpretation: The variance is high because the response time of 300 ms is far from the mean of 150.5 ms.

The value of 300 ms has a deviation of:

$$300 - 150.5 = 149.5 \text{ ms}$$

After squaring:

$$(149.5)^2 = 22,350.25$$

This single value contributes a very large amount to the total variance. Therefore, the unusually high response time has a major effect on the variability of the dataset.

B. Standard Deviation

Definition

The **standard deviation** measures the typical spread of the data values around the mean. It is the square root of the variance.

Formula

$$\text{Standard deviation, } \sigma = \sqrt{\text{Variance}}$$

Calculation

$$\text{Standard deviation} = \sqrt{2,558.15}$$

$$\text{Standard deviation} = 50.58 \text{ ms approximately}$$

Therefore,

$$\text{Standard deviation} = 50.58 \text{ ms}$$

Interpretation

The standard deviation of approximately 50.58 ms indicates that the response times have a noticeable spread around the mean.

The standard deviation is large because one response time is 300 ms, while most of the other response times are between 120 ms and 150 ms.

The high standard deviation suggests that the website does not provide perfectly consistent response times. Most requests are relatively fast, but one request took much longer than the others.

Final Answer for Question 3

- **Variance = 2,558.15 ms²**
- **Standard deviation = 50.58 ms**

4. Determining the Direction of Skewness

Definition of Skewness

Skewness describes whether the distribution of data is symmetrical or whether it is stretched more toward one side.

A dataset may be:

1. Symmetrical
2. Right-skewed or positively skewed
3. Left-skewed or negatively skewed

Analysis of the Given Data

The response times are: 120, 125, 128, 130, 132, 135, 140, 145, 150, 300

Most values are grouped between:

120 ms and 150 ms

However, one value is much larger:

300 ms

This large value extends the distribution toward the right side.

Therefore, the data is:

Right-skewed or positively skewed

Evidence from Mean and Median

The calculated values are:

Mean = 150.5 ms

Median = 133.5 ms

The mean is greater than the median:

Mean > Median

This is a common characteristic of a right-skewed distribution.

The high value of 300 ms pulls the mean toward the right side. The median remains closer to the main group of values.

Therefore,

The data is positively skewed because the extreme value of 300 ms creates a long right tail and causes the mean to be greater than the median.

Effect of the High Response Time

The response time of 300 ms may indicate:

- A temporary server delay
- High website traffic
- A network problem
- A database delay
- A slow request
- A performance issue in the website

The data scientist should investigate this observation to determine whether it is a genuine performance problem or an unusual one-time event.

5. Why the Median May Be More Useful Than the Mean

The **median** may be more useful because the dataset contains an unusually high response time of 300 ms.

The median is:

133.5 ms

The mean is:

150.5 ms

Most response times are between 120 ms and 150 ms. Therefore, the median of 133.5 ms is closer to the typical response time experienced by most users.

Reason 1: The Mean Is Affected by the Extreme Value

The value of 300 ms is much higher than the other response times.

Because the mean uses every value in the calculation, the 300 ms response time increases the average.

The mean becomes:

150.5 ms

This may give the impression that the normal website response time is around 150.5 ms, even though most requests are faster than this.

Reason 2: The Median Is Resistant to Outliers

The median depends on the middle position of the data.

The value of 300 ms is at the end of the ordered dataset. It does not change the fifth and sixth values, which are 132 ms and 135 ms.

Therefore, the median remains:

133.5 ms

The median is less affected by the unusually high response time.

Reason 3: The Median Represents the Typical User Experience

Most users experienced response times between 120 ms and 150 ms.

The median of 133.5 ms is located near the center of this main group.

Therefore, the median provides a more realistic estimate of the typical response time.

Reason 4: The Median Is Better for Skewed Data

The data is right-skewed because of the high value of 300 ms.

For a skewed dataset, the median is often more representative of the central value because it is not strongly influenced by extreme observations.

The mean is useful for calculating the overall average, but the median is more useful when describing the typical website response time.

Overall Conclusion

The website response-time data is:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

The calculated results are:

Statistical Measure	Result
Mean	150.5 ms
Median	133.5 ms
Range	180 ms
Variance	2,558.15 ms ²
Standard deviation	50.58 ms
Direction of skewness	Right-skewed or positively skewed

Statistical Measure	Result
More useful measure for typical response time	Median

The response time of 300 ms is much higher than the other observations. It increases the mean, range, variance, and standard deviation.

The mean response time is 150.5 ms, but the median is 133.5 ms. Since most response times are between 120 ms and 150 ms, the median provides a better representation of the typical website response time.

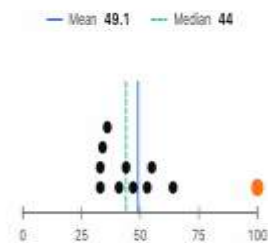
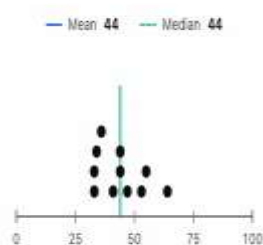
Final conclusion: The data is right-skewed, and the median of 133.5 ms is more useful than the mean for describing the typical response time.

Case Study 3 — Student Performance

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2 and Q3.
2. Calculate the **IQR**.
3. Use the **$1.5 \times \text{IQR}$ rule** to identify any outliers.
4. Find the mean and median.
5. Explain how the outlier affects the mean.



No outlier: mean is 44; median is 44	
No outlier	High outlier
New data set	

Outlier: mean moves 44 → 49.1; median stays 44	
No outlier	High outlier
New data set	

Solution:

CASE STUDY 3: STUDENT PERFORMANCE ANALYSIS

Given Data

The marks obtained by 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

The questions are:

1. Find Q1, Q2, and Q3.
2. Calculate the Interquartile Range (IQR).
3. Use the $1.5 \times \text{IQR}$ rule to identify any outliers.
4. Find the mean and median.
5. Explain how the outlier affects the mean.

1. Calculation of Q1, Q2, and Q3

Definition of Quartiles

Quartiles divide an ordered dataset into four equal parts.

The three main quartiles are:

- **Q1**, or the first quartile, represents the value below which approximately 25% of the observations lie.
- **Q2**, or the second quartile, represents the middle value of the dataset. Q2 is the same as the median.
- **Q3**, or the third quartile, represents the value below which approximately 75% of the observations lie.

The given marks are already arranged in ascending order:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

The number of observations is: **$n = 12$**

Since there are 12 observations, the dataset contains an even number of values.

A. Calculation of Q2

Q2 is the median of the complete dataset.

Since the number of observations is even, the median is calculated by taking the average of the two middle values.

Step 1: Find the middle positions

The two middle positions are:

$$n / 2 = 12 / 2 = 6 \text{ and } (n / 2) + 1 = 6 + 1 = 7$$

Therefore, the sixth and seventh values are used.

Step 2: Identify the sixth and seventh values

Position	Marks
1	42
2	45
3	48

Position	Marks
4	50
5	52
6	52
7	55
8	58
9	60
10	62
11	65
12	95

The sixth value is: 52

The seventh value is: 55

Step 3: Calculate Q2

$$Q2 = (52 + 55) / 2$$

$$Q2 = 107 / 2$$

$$Q2 = 53.5$$

Therefore,

Q2 = 53.5 marks

B. Calculation of Q1

Q1 is the median of the lower half of the dataset.

The lower half contains the first six observations:

42, 45, 48, 50, 52, 52

Since there are six values, the median of this lower half is the average of the third and fourth values.

The third value is: 48

The fourth value is: 50

Calculation

$$Q1 = (48 + 50) / 2$$

$$Q1 = 98 / 2$$

$$Q1 = 49$$

Therefore,

Q1 = 49 marks

Interpretation

Approximately 25% of the students scored at or below 49 marks.

C. Calculation of Q3

Q3 is the median of the upper half of the dataset.

The upper half contains the last six observations:

55, 58, 60, 62, 65, 95

Since there are six values, the median of this group is the average of the third and fourth values.

The third value is: 60

The fourth value is: 62

Calculation

$$Q3 = (60 + 62) / 2$$

$$Q3 = 122 / 2$$

$$Q3 = 61$$

Therefore,

Q3 = 61 marks

Interpretation

Approximately 75% of the students scored at or below 61 marks. The remaining 25% scored above 61 marks.

Final Answer for Question 1

Quartile	Value
Q1	49 marks
Q2 or Median	53.5 marks
Q3	61 marks

Therefore:

Q1 = 49 marks

Q2 = 53.5 marks

Q3 = 61 marks

2. Calculation of the Interquartile Range

Definition

The **Interquartile Range**, commonly called the **IQR**, measures the spread of the middle 50% of the observations.

The IQR is calculated by subtracting Q1 from Q3.

Formula

$$\text{IQR} = \text{Q3} - \text{Q1}$$

From the previous calculations:

$$\text{Q3} = 61$$

$$\text{Q1} = 49$$

Calculation

$$\text{IQR} = 61 - 49$$

$$\text{IQR} = 12$$

Therefore,

IQR = 12 marks

Interpretation of the IQR

The middle 50% of the student marks lie between:

49 marks and 61 marks

The difference between these values is:

$$61 - 49 = 12 \text{ marks}$$

Therefore, the central half of the students has a spread of 12 marks.

The IQR is useful because it focuses on the middle portion of the data. Unlike the range, it is not strongly affected by very high or very low values.

In this dataset, the score of 95 is much higher than the other marks. However, it does not directly affect the calculation of the IQR because the IQR is based only on Q1 and Q3.

Final Answer for Question 2

Interquartile Range = 12 marks

3. Identification of Outliers Using the $1.5 \times \text{IQR}$ Rule

Definition of an Outlier

An **outlier** is an observation that is unusually high or unusually low compared with the rest of the data.

The **$1.5 \times \text{IQR}$ rule** is commonly used to identify possible outliers.

The rule uses two limits:

1. Lower fence
2. Upper fence

Any value below the lower fence or above the upper fence is considered a possible outlier.

Step 1: Calculate $1.5 \times \text{IQR}$

$$\text{IQR} = 12$$

Therefore:

$$1.5 \times \text{IQR} = 1.5 \times 12$$

$$1.5 \times \text{IQR} = 18$$

Step 2: Calculate the Lower Fence

Formula

$$\text{Lower fence} = Q1 - 1.5 \times \text{IQR}$$

Substitute the values:

$$\text{Lower fence} = 49 - 18$$

$$\text{Lower fence} = 31$$

Therefore,

Lower fence = 31 marks

Any score below 31 would be considered a possible lower outlier.

The lowest score in the dataset is 42.

Since: $42 > 31$

there are no lower outliers.

Step 3: Calculate the Upper Fence

Formula

$$\text{Upper fence} = Q3 + 1.5 \times \text{IQR}$$

Substitute the values:

$$\text{Upper fence} = 61 + 18$$

$$\text{Upper fence} = 79$$

Therefore,

Upper fence = 79 marks

Any score above 79 would be considered a possible upper outlier.

The highest score in the dataset is 95.

Since: $95 > 79$

the score of 95 is an outlier.

Final Outlier Result

The acceptable range based on the $1.5 \times \text{IQR}$ rule is:

31 marks to 79 marks

The score of 95 lies above the upper limit.

Therefore, **95 marks is a possible upper outlier.**

There are no lower outliers.

Interpretation

Most student marks are grouped between 42 and 65. The score of 95 is much higher than the other values.

The difference between the highest regular score and the outlier is:

$$95 - 65 = 30 \text{ marks}$$

This large difference shows that the score of 95 is separated from the main group of marks.

The score may represent:

- An exceptionally high-performing student
- A student with strong knowledge of Data Science
- A genuine high score
- An observation that should be checked for data-entry accuracy

An outlier is not necessarily an error. It may be a valid observation that provides useful information about exceptional performance.

4. Calculation of Mean and Median

A. Calculation of Mean

Definition

The **mean** is the arithmetic average of all observations.

Formula

Mean = Sum of all marks / Number of students

The marks are: 42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Step 1: Calculate the total marks

$$\text{Sum} = 42 + 45 + 48 + 50 + 52 + 52 + 55 + 58 + 60 + 62 + 65 + 95$$

$$\text{Sum} = 684$$

Step 2: Find the number of students

Number of students:

$$n = 12$$

Step 3: Calculate the mean

$$\text{Mean} = 684 / 12$$

$$\text{Mean} = 57$$

Therefore,

Mean = 57 marks

Interpretation of the Mean

The average score of the 12 students is 57 marks.

However, the score of 95 increases the total sum and pulls the mean upward.

Most students scored between 42 and 65. The score of 95 is much higher than this main group. Therefore, the mean of 57 is influenced by the high outlier.

B. Calculation of Median

The median is the middle value of an ordered dataset.

The number of observations is:

$$n = 12$$

Since the number is even, the median is the average of the sixth and seventh values.

The sixth value is: 52

The seventh value is: 55

Calculation

$$\text{Median} = (52 + 55) / 2$$

$$\text{Median} = 107 / 2$$

$$\text{Median} = 53.5$$

Therefore,

Median = 53.5 marks

Comparison of Mean and Median

Measure	Value
Mean	57 marks
Median	53.5 marks

Difference:

$$57 - 53.5 = 3.5 \text{ marks}$$

The mean is 3.5 marks higher than the median.

This occurs because the high score of 95 pulls the mean upward.

Final Answer for Question 4

Mean = 57 marks

Median = 53.5 marks

5. Effect of the Outlier on the Mean

The score of 95 is an upper outlier because it is much larger than the other scores.

The outlier affects the mean because the mean includes every value in the calculation.

The formula for the mean is:

Mean = Sum of all observations / Number of observations

Since 95 is included in the total, it increases the sum of the marks.

Mean with the Outlier

Total marks: 684

Number of students: 12

Mean: $684 / 12 = 57$ marks

Therefore, the mean with the outlier is: **57 marks**

Mean Without the Outlier

To understand the effect of the outlier, remove the score of 95.

New total: $684 - 95 = 589$

New number of students: $12 - 1 = 11$

Calculation

Mean without the outlier = $589 / 11$

Mean without the outlier = 53.55 marks approximately

Therefore: **Mean without the outlier $\approx$ 53.55 marks**

Comparison

Calculation	Mean
Mean with the score of 95	57 marks

Calculation	Mean
Mean without the score of 95	53.55 marks

Difference: $57 - 53.55 = 3.45$ marks approximately

Therefore, the score of 95 increases the mean by approximately 3.45 marks.

Explanation

The outlier increases the average because it is much higher than the other values.

Without the outlier, the mean is approximately 53.55 marks.

After including the outlier, the mean becomes 57 marks. Therefore, the high score causes the mean to give a slightly higher estimate of the typical student performance. The median is less affected because it depends on the positions of the middle values. The outlier is located at the upper end of the ordered dataset and does not change the sixth and seventh positions.

Thus:

- The **mean** is affected by the outlier.
- The **median** is more resistant to the outlier.

Overall Conclusion

The student marks are: **42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95**

The final results are:

Statistical Measure	Result
Q1	49 marks
Q2	53.5 marks
Q3	61 marks
IQR	12 marks
Lower fence	31 marks

Statistical Measure	Result
Upper fence	79 marks
Outlier	95 marks
Mean	57 marks
Median	53.5 marks

The middle 50% of the marks lie between 49 and 61. The IQR is 12 marks, showing the spread of the central portion of the data. Using the $1.5 \times \text{IQR}$ rule, the lower limit is 31 and the upper limit is 79. Since 95 is greater than 79, it is identified as a possible upper outlier. The mean is 57 marks, while the median is 53.5 marks. The high score of 95 increases the mean and causes it to be greater than the median.

Final conclusion: The score of 95 is an upper outlier. It increases the mean from approximately 53.55 marks to 57 marks. The median of 53.5 marks is less affected and provides a more stable description of the central student performance.

Case Study 4 — E-Commerce Orders

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Calculate the mean, median, and mode.
2. Find the variance and standard deviation.
3. Calculate the coefficient of variation if the mean and standard deviation are known.
4. What does the standard deviation tell the company?
5. Would you describe the order data as highly or moderately variable?

Solution:

CASE STUDY 4: E-COMMERCE ORDERS ANALYSIS

Given Data

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

The questions are:

1. Calculate the mean, median, and mode.
2. Find the variance and standard deviation.
3. Calculate the coefficient of variation.
4. Explain what the standard deviation tells the company.
5. Determine whether the order data is highly or moderately variable.

1. Calculation of Mean, Median, and Mode

A. Calculation of Mean

Definition

The **mean** is the arithmetic average of all values in a dataset. It is calculated by adding all the observations and dividing the total by the number of observations.

Formula

Mean = Sum of all observations / Number of observations

The order data is:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

Step 1: Calculate the total number of orders

Sum = 18 + 20 + 22 + 22 + 24 + 25 + 26 + 28 + 30 + 35

Sum = 250

Step 2: Find the number of days

Number of observations:

n = 10

Step 3: Apply the mean formula

Mean = 250 / 10

Mean = 25

Therefore,

Mean number of orders = 25 orders per day

Interpretation

The online store received an average of **25 orders per day** during the 10-day period.

The mean gives the company a general estimate of daily demand. It can help the company plan inventory, staff requirements, delivery operations, and customer support.

B. Calculation of Median

Definition

The **median** is the middle value when the observations are arranged in ascending or descending order.

The data is already arranged in ascending order:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

The number of observations is: n = 10

Since the number of observations is even, the median is calculated by taking the average of the two middle values.

Step 1: Find the middle positions

For 10 observations, the middle positions are:

$10 / 2 = 5$ and $(10 / 2) + 1 = 6$

Therefore, the fifth and sixth values are used.

Step 2: Identify the fifth and sixth values

Position	Number of orders
1	18
2	20
3	22
4	22
5	24
6	25
7	26
8	28
9	30
10	35

The fifth value is: 24

The sixth value is: 25

Step 3: Calculate the median

$$\text{Median} = (24 + 25) / 2$$

$$\text{Median} = 49 / 2$$

$$\text{Median} = 24.5$$

Therefore,

Median number of orders = 24.5 orders

Interpretation

The median shows that the central daily order level is approximately **24.5 orders**.

Since the median is close to the mean of 25, the data is reasonably balanced around the center.

C. Calculation of Mode

Definition

The **mode** is the value that occurs most frequently in a dataset.

The order values are:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

The value **22** occurs two times.

All other values occur only once.

Therefore,

Mode = 22 orders

Interpretation

The mode indicates that 22 orders was the most frequently observed daily order count during the 10-day period.

Final Answer for Question 1

Measure	Result
Mean	25 orders
Median	24.5 orders
Mode	22 orders

Therefore:

Mean = 25 orders

Median = 24.5 orders

Mode = 22 orders

2. Calculation of Variance and Standard Deviation

A. Variance

Definition

Variance measures how much the values in a dataset are spread around the mean.

A small variance indicates that the values are close to the mean. A large variance indicates that the values are widely spread.

For this case study, the recorded data is treated as the complete set of 10 days. Therefore, the **population variance formula** is used.

Formula

Variance, $\sigma^2 = \Sigma(x - \mu)^2 / N$

Where:

- x = individual number of orders
- μ = mean number of orders
- N = total number of observations
- Σ = sum of all values

The mean is:

$$\mu = 25$$

Step-by-Step Calculation Table

Orders, x	$x - 25$	$(x - 25)^2$
18	-7	49
20	-5	25
22	-3	9
22	-3	9
24	-1	1
25	0	0
26	1	1
28	3	9
30	5	25
35	10	100

Step 1: Add the squared deviations

$$\Sigma(x - \mu)^2 =$$

$$49 + 25 + 9 + 9 + 1 + 0 + 1 + 9 + 25 + 100$$

$$\Sigma(x - \mu)^2 = 228$$

Step 2: Divide by the number of observations

$$\text{Variance} = 228 / 10$$

$$\text{Variance} = 22.8$$

Therefore,

$$\text{Variance} = 22.8 \text{ orders}^2$$

Interpretation

The variance indicates the overall spread of the daily order values around the mean of 25 orders.

The value of 35 orders contributes more to the variance because it is 10 orders above the mean. When this difference is squared, it becomes:

$$10^2 = 100$$

Therefore, larger deviations have a stronger effect on the variance.

B. Standard Deviation

Definition

The **standard deviation** is the square root of the variance. It measures the typical amount by which the observations differ from the mean.

Formula

Standard deviation, $\sigma = \sqrt{\text{Variance}}$

Calculation

$$\text{Standard deviation} = \sqrt{22.8}$$

$$\text{Standard deviation} \approx 4.77$$

Therefore,

Standard deviation ≈ 4.77 orders

Interpretation

The standard deviation of approximately 4.77 orders means that the daily order count typically varies by about 5 orders from the average of 25 orders.

The order values are not exactly the same every day. However, most of the values remain reasonably close to the mean.

Final Answer for Question 2

Variance = 22.8 orders²

Standard deviation = 4.77 orders

3. Calculation of the Coefficient of Variation

Definition

The **coefficient of variation**, abbreviated as **CV**, measures the standard deviation relative to the mean.

It is expressed as a percentage.

The coefficient of variation is useful when evaluating how much variation exists compared with the average value.

Formula

$$CV = (\text{Standard deviation} / \text{Mean}) \times 100$$

The values are:

$$\text{Mean} = 25$$

$$\text{Standard deviation} = 4.77$$

Calculation

$$CV = (4.77 / 25) \times 100$$

$$CV = 0.1908 \times 100$$

$$CV = 19.08\%$$

Therefore,

$$\text{Coefficient of Variation} \approx 19.08\%$$

Interpretation

A coefficient of variation of approximately 19.08% means that the standard deviation is about 19% of the mean.

This indicates that the daily order counts have a moderate level of relative variation.

The company can expect some changes in daily demand, but the variation is not extremely large.

Final Answer for Question 3

$$\text{Coefficient of Variation} = 19.08\%$$

4. What Does the Standard Deviation Tell the Company?

The standard deviation is approximately:

4.77 orders

The mean is:

25 orders per day

Therefore, the standard deviation tells the company that daily orders generally differ from the average by approximately 4.77 orders.

In simple terms, the company receives about 25 orders per day, but the actual number may vary by around 5 orders on many days.

Practical Interpretation

The company can use the standard deviation for planning.

For example:

- The average number of orders is 25.
- A typical variation is approximately 5 orders.
- Therefore, many daily order values may be expected to fall roughly between 20 and 30 orders.

This is an approximate interpretation:

$$25 - 4.77 \approx 20.23$$

$$25 + 4.77 \approx 29.77$$

Thus, a typical range around one standard deviation from the mean is approximately:

20 to 30 orders

The actual data includes values outside this interval, such as 18 and 35. However, these values help show that daily demand is not perfectly constant.

Importance for Business Planning

The standard deviation can help the company make decisions about:

1. Inventory Management

The company can maintain enough stock to handle normal changes in demand.

Since the average is 25 orders and the standard deviation is approximately 5 orders, inventory planning should consider days with more than 25 orders.

2. Employee Scheduling

The company can schedule enough employees to process the expected number of orders.

On days with higher demand, additional staff may be required.

3. Delivery Planning

The company can estimate the number of deliveries that may need to be completed.

A moderate standard deviation indicates that delivery requirements may change from day to day, but the changes are manageable.

4. Customer Support

More orders may result in more customer questions, payment issues, return requests, or delivery inquiries.

The company can use the order pattern to plan customer-support capacity.

5. Demand Forecasting

The mean provides an estimate of average demand, while the standard deviation indicates the uncertainty around that average.

Using both measures gives a more complete understanding of customer demand.

5. Is the Order Data Highly or Moderately Variable?

The coefficient of variation is:

19.08%

The standard deviation is:

4.77 orders

The mean is:

25 orders

The standard deviation is less than one-fifth of the mean. Therefore, the daily order counts do not show extremely large variation.

The data can be described as:

Moderately variable

Reasons

Reason 1: Most values are close to the mean

The mean is 25 orders.

Most order values are between 20 and 30:

20, 22, 22, 24, 25, 26, 28, and 30

This shows that most daily order counts are reasonably close to the average.

Reason 2: The coefficient of variation is about 19%

The coefficient of variation is below 20%.

This indicates that the relative spread is moderate rather than very high.

Reason 3: The standard deviation is manageable

The standard deviation is approximately 4.77 orders.

Compared with the mean of 25 orders, this is a moderate amount of variation.

Reason 4: There is no extremely unusual value

The highest value is 35 and the lowest value is 18.

Although these values are different from the mean, they are not extremely far from the main group.

Overall Conclusion

The daily order data is:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

The final results are:

Statistical Measure	Result
Mean	25 orders
Median	24.5 orders
Mode	22 orders
Variance	22.8 orders ²
Standard deviation	4.77 orders
Coefficient of variation	19.08%
Level of variability	Moderate

The mean of 25 orders indicates the average daily demand. The median of 24.5 orders is close to the mean, showing that the data is reasonably balanced around the center. The mode of 22 orders indicates the most frequently observed order count. The standard deviation of approximately 4.77 orders shows that daily orders generally vary by about 5 orders from the average. The coefficient of variation of approximately 19.08% indicates a moderate level of relative variation.

Final conclusion: The e-commerce order data is moderately variable. The company receives an average of 25 orders per day, with a typical variation of approximately 5 orders. This information can help the company plan inventory, staffing, delivery operations, and customer support.

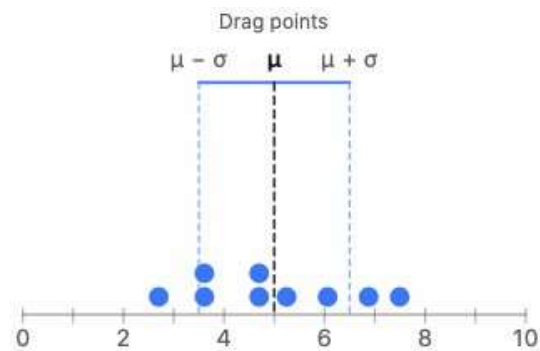
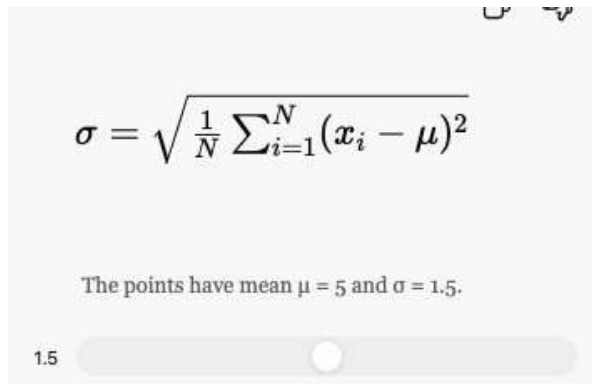
Case Study 5 — Comparing Two Data Science Models

Two machine-learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the mean error for both models.
2. Calculate the variance for both models.
3. Calculate the standard deviation for both.
4. Which model has more consistent predictions?
5. Can the model with the lower mean error necessarily be considered better? Explain.



Solution:

CASE STUDY 5: COMPARING TWO DATA SCIENCE MODELS

Given Data

Two machine-learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

The questions are:

1. Calculate the mean error for both models.
2. Calculate the variance for both models.
3. Calculate the standard deviation for both models.
4. Identify which model has more consistent predictions.
5. Explain whether the model with the lower mean error can always be considered better.

For the variance calculations, the given values are treated as the complete datasets.
Therefore, the **population variance formula** is used.

1. Calculation of Mean Error

Definition

The **mean error** is the average prediction error produced by a model.

It is calculated using the formula:

Mean = Sum of all error values / Number of error values

A lower mean error generally indicates that the model's predictions are closer to the actual values on average.

However, mean error alone does not explain how consistently the model performs.

Therefore, variance and standard deviation must also be considered.

A. Mean Error of Model A

The prediction errors of Model A are: 2, 3, 3, 4, 4, 5, 5, 6

Step 1: Find the sum of all errors

$$\text{Sum} = 2 + 3 + 3 + 4 + 4 + 5 + 5 + 6$$

$$\text{Sum} = 32$$

Step 2: Find the number of observations

Number of error values:

$$n = 8$$

Step 3: Apply the mean formula

$$\text{Mean error} = 32 / 8$$

$$\text{Mean error} = 4$$

Therefore,

Mean error of Model A = 4

Interpretation

Model A produces an average prediction error of 4 units. The error values are grouped closely around the mean. Most of the errors lie between 2 and 6.

B. Mean Error of Model B

The prediction errors of Model B are: 1, 1, 2, 4, 5, 7, 8, 9

Step 1: Find the sum of all errors

$$\text{Sum} = 1 + 1 + 2 + 4 + 5 + 7 + 8 + 9$$

$$\text{Sum} = 37$$

Step 2: Find the number of observations

Number of error values:

$$n = 8$$

Step 3: Apply the mean formula

$$\text{Mean error} = 37 / 8$$

$$\text{Mean error} = 4.625$$

Therefore,

$$\text{Mean error of Model B} = 4.625$$

Interpretation

Model B produces an average prediction error of approximately 4.63 units.

The error values of Model B are spread over a wider range, from 1 to 9.

Comparison of Mean Errors

Model	Mean Error
Model A	4.00
Model B	4.625

Model A has the lower mean error.

Difference:

$$4.625 - 4.00 = 0.625$$

Therefore, Model A has an average error that is **0.625 units lower** than Model B.

Final Answer for Question 1

$$\text{Mean error of Model A} = 4$$

$$\text{Mean error of Model B} = 4.625$$

Model A has the lower mean error.

2. Calculation of Variance

Definition

Variance measures how widely the error values are spread around the mean.

A smaller variance indicates that the errors are closer to the mean and the model performs more consistently.

A larger variance indicates that the errors are more widely spread and the model's performance is less consistent.

Population Variance Formula

$$\sigma^2 = \Sigma(x - \mu)^2 / N$$

Where:

- x = individual error value
- μ = mean error
- N = total number of error values
- σ^2 = population variance

A. Variance of Model A

The mean error of Model A is:

$$\mu = 4$$

The error values are:

2, 3, 3, 4, 4, 5, 5, 6

Step-by-Step Calculation Table

Error, x	$x - 4$	$(x - 4)^2$
2	-2	4
3	-1	1
3	-1	1
4	0	0
4	0	0
5	1	1
5	1	1
6	2	4

Step 1: Find the sum of squared deviations

$$\Sigma(x - \mu)^2 = 4 + 1 + 1 + 0 + 0 + 1 + 1 + 4$$

$$\Sigma(x - \mu)^2 = 12$$

Step 2: Divide by the number of observations

$$\text{Variance} = 12 / 8$$

$$\text{Variance} = 1.5$$

Therefore,

$$\text{Variance of Model A} = 1.5 \text{ error units}^2$$

Interpretation

The variance of 1.5 is relatively small. This shows that the errors of Model A remain close to the mean error of 4.

B. Variance of Model B

The mean error of Model B is:

$$\mu = 4.625$$

The error values are: 1, 1, 2, 4, 5, 7, 8, 9

Step-by-Step Calculation Table

Error, x	$x - 4.625$	$(x - 4.625)^2$
1	-3.625	13.140625
1	-3.625	13.140625
2	-2.625	6.890625
4	-0.625	0.390625
5	0.375	0.140625
7	2.375	5.640625
8	3.375	11.390625
9	4.375	19.140625

Step 1: Find the sum of squared deviations

$$\Sigma(x - \mu)^2 = 69.875$$

Step 2: Divide by the number of observations

$$\text{Variance} = 69.875 / 8$$

$$\text{Variance} = 8.734375$$

Therefore,

$$\text{Variance of Model B} \approx 8.73 \text{ error units}^2$$

Interpretation

The variance of Model B is much larger than the variance of Model A.

This means that the errors of Model B are more widely spread around the mean.

Model B has some very low errors, such as 1 and 2, but it also has high errors, such as 8 and 9. This creates greater variation.

Comparison of Variance

Model	Variance
Model A	1.50
Model B	8.73

Model B has a much larger variance.

Therefore, Model A has more stable and less variable prediction errors.

Final Answer for Question 2

Variance of Model A = 1.50

Variance of Model B = 8.73

Model A has lower variance.

3. Calculation of Standard Deviation

Definition:

The **standard deviation** is the square root of the variance.

It measures the typical spread of the prediction errors around the mean.

Formula: Standard Deviation = $\sqrt{\text{Variance}}$

A smaller standard deviation indicates more consistent predictions.

A larger standard deviation indicates greater variation in prediction errors.

A. Standard Deviation of Model A

Variance of Model A: $\sigma^2 = 1.5$

Calculation

Standard deviation = $\sqrt{1.5}$

Standard deviation ≈ 1.2247

Therefore,

Standard deviation of Model A ≈ 1.22

Interpretation

The prediction errors of Model A typically differ from the mean by approximately 1.22 units. The errors are closely grouped around the mean error of 4.

B. Standard Deviation of Model B

Variance of Model B: $\sigma^2 = 8.734375$

Calculation

Standard deviation = $\sqrt{8.734375}$

Standard deviation ≈ 2.955

Therefore,

Standard deviation of Model B ≈ 2.96

Interpretation

The prediction errors of Model B typically differ from the mean by approximately 2.96 units. The larger standard deviation indicates that Model B's errors are more spread out.

Comparison of Standard Deviation

Model	Standard Deviation
Model A	1.22
Model B	2.96

Model B has a standard deviation that is much larger than that of Model A.

Therefore, Model A has more consistent prediction errors.

Final Answer for Question 3

Standard deviation of Model A ≈ 1.22

Standard deviation of Model B ≈ 2.96

4. Which Model Has More Consistent Predictions?

Consistency refers to how stable and predictable the prediction errors are.

A model is considered more consistent when its errors have:

- Lower variance
- Lower standard deviation
- Less spread around the mean

The results are:

Measure	Model A	Model B
Mean error	4.00	4.625
Variance	1.50	8.73
Standard deviation	1.22	2.96

Model A has:

- A lower mean error
- A lower variance
- A lower standard deviation

Therefore,

Model A has more consistent predictions.

Explanation

The errors of Model A range from 2 to 6.

The errors of Model B range from 1 to 9.

Although Model B produces some very low errors, it also produces some very high errors.

For example:

- Model B has errors of 1 and 2, which are low.
- Model B also has errors of 8 and 9, which are high.

This large spread makes Model B less consistent.

Model A does not have extremely low or extremely high errors. Its errors remain close to the mean.

Therefore, Model A gives more stable performance.

5. Can the Model with the Lower Mean Error Always Be Considered Better?

No. A model with a lower mean error cannot always be considered better.

Mean error is important, but it is only one measure of model performance.

A complete evaluation should also consider:

- Variance
- Standard deviation
- Consistency
- Maximum error
- Performance on different types of data
- The importance of large errors
- The application in which the model is used

Reason 1: Mean Does Not Show the Spread of Errors

Two models may have similar mean errors but very different levels of variation.

For example:

- One model may produce errors that are always close to 4.
- Another model may produce some errors close to 0 and some errors close to 10.

Both models could have similar average errors, but their reliability would be different.

Therefore, the mean alone does not describe consistency.

Reason 2: Large Errors May Be Important

In some applications, a few very large errors can cause serious problems.

For example:

- In medical prediction, a large error may affect a clinical decision.
- In financial forecasting, a large error may result in a major financial loss.
- In weather prediction, a large error may affect safety planning.
- In autonomous systems, a large error may create a safety risk.

Therefore, a model with a low mean error may still be unsuitable if it occasionally produces very large errors.

Reason 3: Consistency Is Important

A model with slightly higher mean error but very stable performance may be preferred in some situations.

For example, consider two hypothetical models:

Model X: Errors = 4, 4, 4, 4, 4 & Mean error = 4

Model Y: Errors = 0, 1, 2, 7, 10 & Mean error = 4

Both models have the same mean error.

However, Model X is more consistent because all its errors are equal.

Model Y has highly variable errors.

Therefore, mean error alone is not enough to determine which model is better.

Reason 4: The Choice Depends on the Application

The best model depends on the purpose of the system.

If consistent performance is important, a model with lower variance may be preferred.

If achieving the lowest possible average error is the main goal, the model with the lower mean error may be preferred.

The decision should be based on both average accuracy and reliability.

Overall Comparison

Statistical Measure	Model A	Model B
Mean error	4.00	4.625
Variance	1.50	8.73
Standard deviation	1.22	2.96
Error range	2 to 6	1 to 9
Consistency	Higher	Lower

Overall Conclusion

The prediction errors are:

Model A:

2, 3, 3, 4, 4, 5, 5, 6

Model B:

1, 1, 2, 4, 5, 7, 8, 9

The final results are:

- **Mean error of Model A = 4**
- **Mean error of Model B = 4.625**
- **Variance of Model A = 1.50**
- **Variance of Model B ≈ 8.73**
- **Standard deviation of Model A ≈ 1.22**
- **Standard deviation of Model B ≈ 2.96**

Model A has both a lower average error and lower variability.

Therefore:

Model A has more consistent predictions and performs better for this particular dataset.

However, a lower mean error alone is not always sufficient to determine that a model is better. Variance, standard deviation, large errors, reliability, and application requirements must also be considered.

Final conclusion: Model A is the more consistent model because its prediction errors are closely grouped around the mean. Model B has greater variation and is less predictable.